

Why HCI?

Aris Alissandrakis

Learning Goals

- Recognise the role of the user interface in computing systems
- Know basic expectations users have of technology

Intel XP/S 140 Paragon

- 3,680 processor cores
- 143.40 GFLOPS
- Fastest computer in the world (June 1994)





Valve SteamDeck

- 4 processor cores
- 1,600 GFLOPS
- Fastest computer on Aris' sofa (Oct 2024)

- Computing power is not a limit anymore
- Interface determines the form factor (size, shape, etc)
- We build tools that extend our abilities

But ... WHAT IS a computer?

Computation (?)

Components (?)

Usage (?)

So, why this course?

We build artefacts with which people will **interact**.

These artefacts inhabit a **human** environment.

We learn how to design things that **fit the environment**.

And **verify** if they artefacts are doing what they are supposed to do.

"[HCI] is the discipline focused on understanding of the way people use and are impacted by computer technology and the way that technology can be designed to better serve human needs. This includes single users interacting directly with devices, but also the impact of computers at the level of work groups, families, organisations and communities." ([Alan Dix](#))

Digital vs. Mechanical

- Affordance
- Visibility



Is it difficult to create user interfaces?

Consider sketching an easy to use interface for an alarm clock ...

- ... on a mobile phone platform
- ... as a physical device
- ... as an application for a TV

Is there room for improvement?

What questions do you need to ask?

What decisions will you make – implicitly / explicitly?